Bogleheads

Investment risk management

Investment risk management discusses management of risk in a portfolio.

Definition of risk

There are two dimensions to risk: probability of failure; and magnitude of failure

Risk is the product of those two dimensions:[1]

Risk = (probability of failure) × (magnitude of failure)

For example, two portfolios have a real withdrawal amount of $30,000 per year for 30 years.[2]

- Portfolio A: 70% stocks /30% bonds, with a 4% failure rate at the given withdrawal rate
- Portfolio B: 30% stocks /70% bonds, with a 6% failure rate at the given withdrawal rate

The total risk *value* is therefore:

- Portfolio A = .04 × $190,000 = $7,600
- Portfolio B = .06 × $100,000 = $6,000

Portfolio B is less risky because it avoids the big losses in the early years that sometimes devastate Portfolio A. For example, a 50% stock market decline in first three years of retirement.

Risk is also defined as the chance that an investment's actual return will be different than expected. This includes the possibility of losing some or all of the original investment. Risk related to an investment is often called *investment risk*.[3]

Risk transfer techniques

Much financial risk is handled by risk transfer techniques.[4] The three risk transfer techniques are hedging, insuring, and diversifying.

- **Hedging risk:** A hedging risk technique means giving up upside potential in return for locking in a known value. For example, when you purchase a life annuity you are hedging retirement income risk.
- **Insuring risk:** An insuring risk technique means paying a premium to ensure a floor, but, unlike hedging, keep some upside potential.
- **Diversifying risk:** A diversifying risk technique does not lock in anything, but you don't pay a premium and you have unlimited upside potential.

Risk strategies that rely on hedging and insuring are called *matching strategies* because they match assets to liabilities. Risk strategies based on diversifying are simply called diversification strategies.

Most Bogleheads handle investment risk solely through diversification strategies. The minority of Bogleheads that use matching strategies typically rely on all three risk transfer techniques, but emphasize hedging and insuring over diversifying when handling investment risk.

For many households, the two main financial goals are paying for a child's college education and generating enough income in retirement to meet their retirement living standard goal. Here are examples between the two schools of risk management on meeting those two primary financial goals.

Saving and investing for a child's college education

The *diversification-only* risk management Bogleheads save and invest in 529 college cost plans and are concerned about which plans offer the best combination of risky and low risk mutual funds at low cost.

The *matching strategy first* risk management Bogleheads emphasize pre-paid tuition plans, TIPS, and I-bonds when saving and investing for college costs. While they may use some equity mutual funds in the investment plan, such mutual fund products are secondary in the overall college cost investment plan.

Saving and investing for retirement

The *diversification-only* risk management Bogleheads set an asset allocation strategy in their retirement portfolio between risky and less risky assets that they believe will maximize their expected return given the amount of portfolio volatility they are willing to tolerate. They then rebalance to this asset allocation over time, or rebalance the asset allocation and also adjust the asset allocation for age over time.

The *matching strategy first* Bogleheads set an annual income target for retirement. They then decide how much of that income target should be met by a combination of relatively safe matching strategy assets such as Social Security, defined benefit pensions, life annuities, and TIPS ladders. In pre-retirement they set aside a part of their portfolio in safe assets such as TIPS to hedge and ensure that they will reach the level of assets they will need at retirement to purchase safe retirement income assets, for example, a life annuity, to meet their retirement matching income sub-target. The remainder of the retirement portfolio uses diversification strategies among risky and less risky investment assets in an effort to fund retirement income above the matching income sub-target.

Matching strategies using hedging and insuring risk management techniques control downside financial risk much more tightly than diversification strategies, but at the cost of giving up most or all upside potential above the target level of financial assets required to meet a financial goal. Diversification strategies offer unlimited upside potential, but much less downside risk control compared to matching strategies. Bogleheads fundamentally disagree about how to manage this risk/return tradeoff.

Portfolio return risk statistics

Portfolio returns are expected to follow a normal distribution. Deviations from this pattern are a form of investment risk.[3][5]

References

1. Bogleheads forum post: "Redefining risk" (https://www.bogleheads.org/forum/viewtopic.php?p=1162119#p1162119), bobcat2. August 26, 2011.
2. Bogleheads forum post: "Redefining risk" (https://www.bogleheads.org/forum/viewtopic.php?p=1163546#p1163546), bobcat2. August 27, 2011.
3. Risk - Financial risk, from Wikipedia.
4. Bogleheads forum post: "What do bogleheads *not* agree on?" (https://www.bogleheads.org/forum/viewtopic.php?p=964361#p964361), bobcat2. February 19, 2011.
5. Financial risk, from Wikipedia.

External links

- Mismeasurement of risk in financial planning (https://web.archive.org/web/20160410111203/http://www.plansponsor.com/uploadfiles/russell.pdf), Richard K. Fullmer, CFA, October 2009. Archived from the original (http://www.plansponsor.com/uploadfiles/russell.pdf) on April 10, 2016.
- Risk: What It Means in Investing and How to Measure and Manage It (https://www.investopedia.com/terms/r/risk.asp), from Investopedia.

Forum discussions

- Bogleheads forum topic: "Redefining risk" (https://www.bogleheads.org/forum/viewtopic.php?t=81156)
- Bogleheads forum topic: "What do bogleheads *not* agree on?" (https://www.bogleheads.org/forum/viewtopic.php?t=69104)
- Bogleheads forum topic: "Understanding all the talk about "tails" " (https://www.bogleheads.org/forum/viewtopic.php?t=69213)
- Bogleheads forum topic: "Risk - A Primer" (https://www.bogleheads.org/forum/viewtopic.php?t=83960)

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